APAARSetu

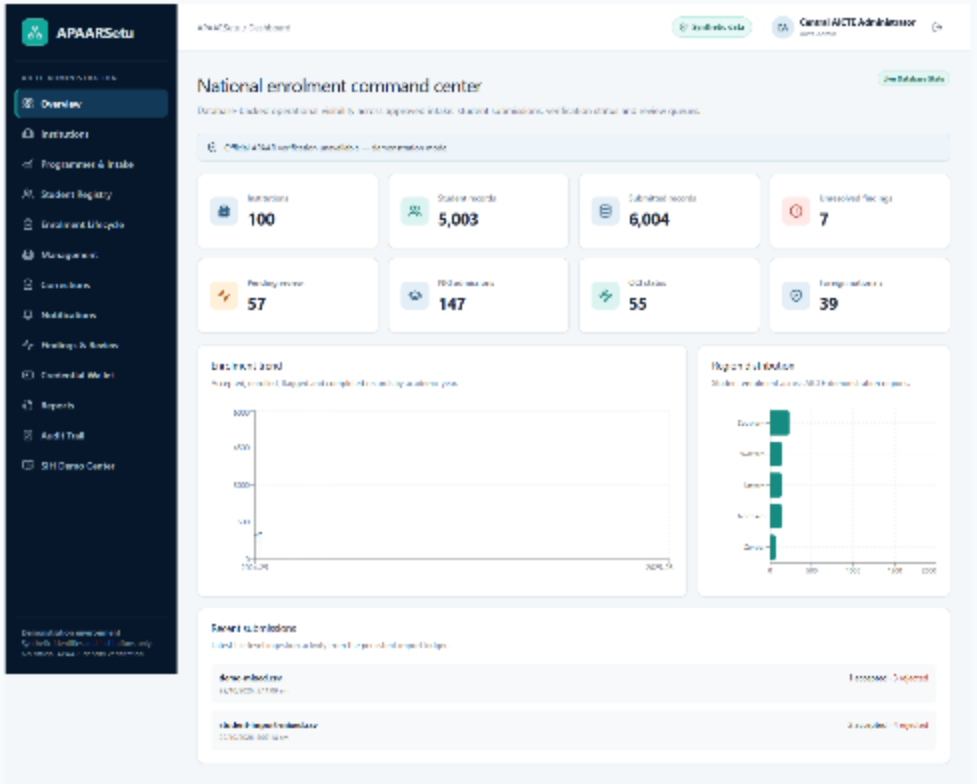
Intelligent student data management, enrolment compliance and academic verification

SIH26263
Smart India Hackathon 2026
Synthetic demonstration

Proposed solution and uniqueness

One repository connects institutional submissions with accountable review

Identify duplicate and inconsistent records
Apply centrally authorized intake policies
Explain every finding with stored evidence
Let students control credential disclosure



Technical approach

Four portals share a versioned API and persistent records

Next.js and strict TypeScript
FastAPI, SQLAlchemy and Alembic
SQLite locally with relational constraints
Rule validation and Isolation Forest
Ed25519 signatures with scoped consent

Feasibility and viability

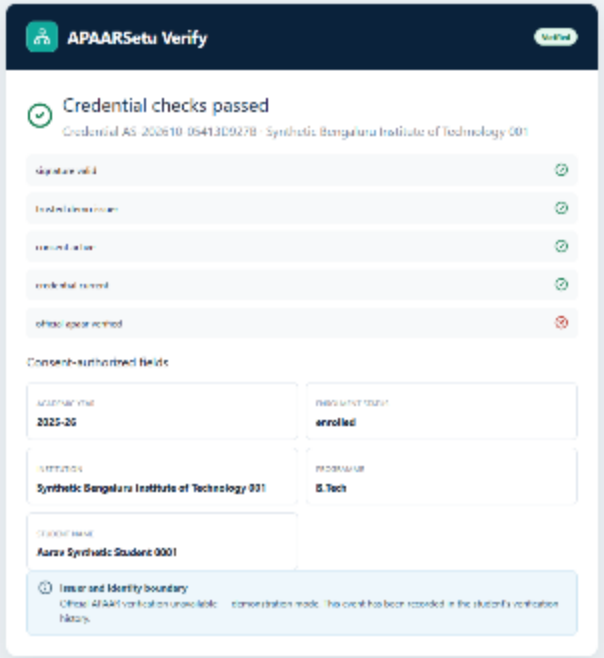
Native Windows operation without paid AI services

100 fictional institutions and 5,000 identities
CSV/XLSX preview and transactional ingestion
Automated authorization and signature tests
Synthetic hold-out precision 0.80 and recall 1.00
Production requires approved integrations and security assessment

Impact and expected benefits

Administrative visibility with student-controlled disclosure

Earlier visibility of intake discrepancies
Traceable correction and review decisions
Limited disclosure to verification recipients
Measured workflow evidence from synthetic data



Research and references

Official identity evidence remains a separate trust boundary

APAAR FAQ: apaar.education.gov.in/faqs
Isolation Forest: scikit-learn.org
Ed25519: cryptography.io
FastAPI and SQLAlchemy documentation
Project evaluation and actual application screenshots
Official APAAR verification unavailable in demonstration mode